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1. What is the primary difference between "Quantity Survey" and "Quantity Estimation" as defined in the text?

a) Survey is for materials, estimation is for labor.

b) Survey is an accurate calculation from maps/site, while estimation is an approximate assessment without site measurements.

c) Survey is done by the contractor, estimation by the engineer.

d) There is no difference; they are the same process.

Answer: b) Survey is an accurate calculation from maps/site, while estimation is an approximate assessment without site measurements.

2. According to the document, which of the following is NOT a requirement for quantity estimation?**

a) Having full knowledge of the work to be estimated.

b) Having a fully informed knowledge about the position of the project.

c) Having a final, signed contract with the client.

d) Having the ability to assess the essential steps of work progress.

Answer: c) Having a final, signed contract with the client.

3. During the "Planning Stage" of a project, what is the primary role of the estimator?**

a) To prepare the final bill of quantities.

b) To provide preliminary cost estimations based on preliminary plans.

c) To supervise the construction work on site.

d) To negotiate the final contract with the owner.

Answer: b) To provide preliminary cost estimations based on preliminary plans.

4. One of the benefits of cost estimation during the planning stage is:**

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a) To determine the exact number of bricks needed.

b) To compare between a number of alternatives.

c) To sign the final work contract.

d) To order all materials for construction.

Answer: b) To compare between a number of alternatives.

5. During the "Stage of preparing contracting documents and contracting", what is a key duty of the estimator?**

a) Performing the physical excavation work.

b) Adjusting and following-up the cost, and reviewing estimates.

c) Hiring the labor force for the project.

d) Laying the foundation for the building.

Answer: b) Adjusting and following-up the cost, and reviewing estimates.

6. Which of the following is a duty of a quantity surveyor during the "construction and delivery stage"?**

a) Developing preliminary plans.

b) Preparing final measures for the constructed works.

c) Conducting feasibility studies for the project.

d) Designing the architectural layout.

Answer: b) Preparing final measures for the constructed works.

7. What is the underlying principle of the "Square Meter Method" for preliminary estimation?**

a) It calculates cost based on the volume of the building.

b) It estimates the cost per cubic meter based on previous projects.

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c) It calculates the cost of a new building based on the cost per square meter of previously constructed buildings.

d) It estimates cost by calculating the surface area of all walls.

Answer: c) It calculates the cost of a new building based on the cost per square meter of previously constructed buildings.

8. Why is the "Cubic Method" considered more accurate than the Square Meter Method?**

a) It only accounts for the ground floor area.

b) It takes into account the rise (height) of the building.

c) It is a newer method with more complex calculations.

d) It disregards the cost of the basement and roof.

Answer: b) It takes into account the rise (height) of the building.

9. The text states that the "Surface-Area Method" can be inaccurate for multi-story buildings because:**

a) It overestimates the cost of the foundation.

b) Adding a floor does not just add horizontal space, but also adds side walls regardless of the number of openings.

c) It only considers the cost of the roof.

d) It is too difficult to calculate the surface area.

Answer: b) Adding a floor does not just add horizontal space, but also adds side walls regardless of the number of openings.

10. The "Story Enclosure Method" is considered more accurate because it:**

a) Only calculates the area of the ground floor.

b) Takes into account the different costs of floors by creating a hypothetical area based on the effect of all basic components.

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c) Ignores the effect of exterior walls on the cost.

d) Uses a simple average of all floor costs.

Answer: b) Takes into account the different costs of floors by creating a hypothetical area based on the effect of all basic components.

11. What is the first step in the "Steps of Quantity Surveying Procedure"?**

a) Abstracting

b) Bill of Quantities

c) Taking off

d) Squaring Dimensions

Answer: c) Taking off

12. What does the "Squaring Dimensions" step in quantity surveying primarily involve?**

a) Taking measurements from the site.

b) Performing calculations, mostly addition and multiplication, on the taken dimensions.

c) Grouping similar work items together.

d) Preparing the final price list.

Answer: b) Performing calculations, mostly addition and multiplication, on the taken dimensions.

13. What is the purpose of the "Abstracting" step in the quantity surveying procedure?**

a) To write down the dimensions from the plans.

b) To calculate the square root of complex areas.

c) To compile similar paragraphs of work into a single step.

d) To determine the unit price of materials.

Answer: c) To compile similar paragraphs of work into a single step.

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14. When preparing a Bill of Quantities (BoQ), what should be done with similar work paragraphs?**

a) They should be listed separately to avoid confusion.

b) They should be assembled under unified (common) titles.

c) They should be omitted from the final bill.

d) They should be replaced with a lump sum item.

Answer: b) They should be assembled under unified (common) titles.

15. The "Method of Construction Items" for taking off involves estimating by:**

a) Grouping all structural elements together.

b) Dividing the project into key elements like foundations, columns, and walls.

c) Taking the sequence in which the project is carried out (e.g., excavation first, then concrete).

d) Estimating all finishes before structural work.

**Answer: c) Taking the sequence in which the project is carried out (e.g., excavation first, then concrete).

16. Which method of taking off is described as the "latest style" and best suited for complex buildings by dividing the project into key elements?**

a) Method of Construction Items

b) Method of Structural Elements

c) Centre Line Method

d) Cross Section Method

Answer: b) Method of Structural Elements

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17. * Which standard unit of measurement (Lump Sum) is used for works that are difficult to measure by conventional methods?**

a) Square meter (m^2)

b) Cubic meter (m^3)

c) Meter length (L. m)

d) Lump Sum (L.S.)

Answer: d) Lump Sum (L.S.)

18. In which type of work is the "Square Meter (m^2)" most appropriately used?**

a) Excavation for foundations

b) Reinforced concrete works

c) Plastering and painting

d) Steel reinforcement

Answer: c) Plastering and painting

19. The "Cubic meter (m^3)" is used to express work where:**

a) Only one dimension is important.

b) The third dimension is very small and hard to measure.

c) All three dimensions are clear and can be measured accurately.

d) The work has no dimensions at all.

Answer: c) All three dimensions are clear and can be measured accurately.

20. "Meter length (L. m)" as a unit is used for works like:**

a) Concrete for foundations.

b) Sanitary works and electrical installation.

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c) Mosaic tiles for flooring.

d) Points of light fixtures.

Answer: b) Sanitary works and electrical installation.

21. What is the standard unit for measuring reinforcement works (rebar)?**

a) Cubic meter (m^3)

b) Square meter (m^2)

c) Number (No.)

d) Weight (Wt.)

Answer: d) Weight (Wt.)

22. **22. According to the standard units, which of the following is measured by the "Number (No.)"?**

a) Site planning

b) Skirting tiles

c) Sanitary works like a washbasin

d) Earth filling

Answer: c) Sanitary works like a washbasin

23. How is "Drilling footing excavation" typically measured in a Bill of Quantities?**

a) By Lump Sum (L.S.)

b) By Square meter (m^2)

c) By Cubic meter (m^3)

d) By Meter length (L. m)

Answer: c) By Cubic meter (m^3)

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24. A "Blinding layer" for footing is measured by:**

a) m^3

b) m^2 with mention of material and thickness

c) L. m

d) Lump Sum

Answer: b) m^2 with mention of material and thickness

25. How is a "Damp Proof Course (D.P.C.)" typically measured?**

a) By m^3 only

b) By m^2 with thickness or m.L with thickness and width

c) By number (No.)

d) By weight (Wt.)

Answer: b) By m^2 with thickness or m.L with thickness and width

26. **26. In the example for calculating wall dimensions under the D.P.C. level, what does the variable "X" represent?**

a) The total height of the wall.

b) The difference in width between the footing and the wall, divided by two. (e.g., $(w_0-w_1)/2$)

c) The length of the centre line.

d) The thickness of the mortar.

Answer: b) The difference in width between the footing and the wall, divided by two. (e.g., $(w_0-w_1)/2$)

27. In the provided example, why are there different lengths for the bricks work at each step (0.48m, 0.36m, 0.24m)?**

a) Because the wall height changes at each step.

b) Because the length of the centre line changes for each different width of the brick layer.

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c) Because the material for bricks changes at each step.

d) Because the cost of bricks is different at each level.

Answer: b) Because the length of the centre line changes for each different width of the brick layer.

28. For a standard brick with dimensions (7 x 11 x 23) cm, what is its approximate volume after building with 1cm of mortar?**

a) 0.00177 m^3

b) 0.0023 m^3

c) 0.0192 m^3

d) 0.0288 m^3

Answer: b) 0.0023 m^3

29. Approximately how many bricks are needed for one cubic meter of brickwork (after accounting for mortar)?**

a) 52 bricks

b) 35 bricks

c) 435 bricks

d) 500 bricks

Answer: c) 435 bricks

30. For a wall with a 12 cm thickness, approximately how many bricks are needed for one square meter of wall face?**

a) 35 bricks

b) 52 bricks

c) 435 bricks

d) 500 bricks

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Answer: b) 52 bricks

31. For a wall with an 8 cm thickness, approximately how many bricks are needed for one square meter of wall face?**

a) 35 bricks

b) 52 bricks

c) 435 bricks

d) 500 bricks

Answer: a) 35 bricks

32. What is the approximate volume of mortar required for 1 m^3 of brickwork?**

a) 0.77 m^3

b) 0.23 m^3

c) 0.02 m^3

d) 1.00 m^3

Answer: b) 0.23 m^3

33. If the volume of cement mortar after use (V) is given by V = 0.75 (C + S), what does the 0.75 factor represent?**

a) The weight of cement per m^3.

b) The reduction in volume due to the sand filling the voids in the cement.

c) The number of bricks in the wall.

d) The thickness of the plaster.

Answer: b) The reduction in volume due to the sand filling the voids in the cement.

34. For a 2 cm thick plaster on 1 m^2 of wall, what is the volume of mortar needed?**

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a) 0.02 m^3

b) 0.2 m^3

c) 2 m^3

d) 0.002 m^3

Answer: a) 0.02 m^3

35. For a 1 m^3 concrete mix, the formula V = 0.67 (C + S + A) is used. What does the 0.67 factor account for?**

a) The weight of the cement.

b) The volume of water in the mix.

c) The reduction in volume as sand and gravel combine and fill voids.

d) The amount of reinforcement in the concrete.

Answer: c) The reduction in volume as sand and gravel combine and fill voids.

36. Using the formula for concrete volume, if the mix rate is (1:2:4), what is the proportion of gravel?**

a) 1

b) 2

c) 4

d) 8

Answer: c) 4

37. In the example for a (1:2:4) concrete mix, if 10 m^3 of concrete is required, how many 50 kg bags of cement are approximately needed?**

a) 1610 bags

b) 33 bags

c) 50 bags

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d) 115 bags

Answer: b) 33 bags (1610 kg / 50 kg = 32.2 ≈ 33 bags)

38. What is the typical percentage range for the reduction in volume when sand and gravel are mixed?**

a) 1% to 5%

b) 5% to 10%

c) 15% to 20%

d) 40% to 50%

Answer: c) 15% to 20%

39. When calculating the number of (20x20) cm tiles needed for a floor, what allowance is made for the joints between tiles?**

a) 1 mm

b) 2 mm

c) 3 mm

d) No allowance is needed.

Answer: c) 3 mm

40. How many (20x20) cm tiles, including the 3mm joint, are approximately needed for 1 m^2 of flooring?**

a) 20

b) 24.3 (approx. 24)

c) 35

d) 52

Answer: b) 24.3 (approx. 24)

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41. In the context of the "Story Enclosure Method", which factor is multiplied by 1 to find the hypothetical building space?**

a) Ground floor area

b) Area of the first floor

c) Ceiling space

d) All of the above

Answer: c) Ceiling space

42. According to the document, what is the role of an estimator in the "Preparation of Variation Orders"?**

a) To design the variation.

b) To price the additional works.

c) To approve the variation on site.

d) To ignore the variation.

Answer: b) To price the additional works.

43. In the Bill of Quantities example for the room, how was the length for the excavation works calculated?**

a) By using the external perimeter of the building.

b) By calculating the centre line length of the foundation trench.

c) By measuring the longest wall.

d) By using the area of the room.

Answer: b) By calculating the centre line length of the foundation trench.

44. In the same BoQ example, why is the volume of the lintel beam subtracted from the volume of brickwork above the D.P.C.?**

a) Because the lintel is made of a different type of brick.

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b) Because the space occupied by the lintel is not filled with brickwork.

c) Because the lintel is not part of the wall.

d) Because the lintel is cheaper than bricks.

Answer: b) Because the space occupied by the lintel is not filled with brickwork.

45. What is the specified thickness of the asphalt layer for roofing works in the BoQ example?**

a) 0.02 m

b) 0.1 m

c) 0.05 m

d) 0.15 m

Answer: a) 0.02 m

46. For which type of works is the use of "resistant salt type of cement" specified in the BoQ example?**

a) All concrete works

b) Only for the roof slab

c) All underground works

d) Only for the plastering

Answer: c) All underground works

47. In the same example, what is the width used for plastering the internal edges of doors and windows?**

a) 0.24 m

b) 0.12 m

c) 0.10 m

d) 0.05 m

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Answer: c) 0.10 m

48. For a hexagonal-shaped building plan in the homework section, what fundamental calculation is needed to find the length of the centre line?**

a) Adding the lengths of all exterior walls.

b) Using trigonometric functions to find the dimensions of the component triangles.

c) Calculating the total area of the hexagon.

d) Multiplying the number of sides by the width of a single wall.

Answer: b) Using trigonometric functions to find the dimensions of the component triangles.

49. According to the section on "Construction Material Quantities," what is the weight of 1 m^3 of gypsum (including added water)?**

a) 1400 kg

b) 1275 kg

c) 50 kg

d) 28.3 kg

50. In the "Quantity of Cement & Sand required for Finishing" section, what percentage of gypsum volume is stated to be lost during application?**

a) 5%

b) 10%

c) 15%

d) 20%

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